

July 7th, 2026

Today I replaced several simulated telemetry values with live sensor data. I integrated a battery voltage sensor by tapping into the battery-to-ESC power leads and routing the voltage sensor output to the ESP32. The breadboard was used to distribute 3.3V and GND to all sensors while keeping each sensor's signal wire connected to its own GPIO pin.

I also integrated a DS18B20 temperature sensor for live motor temperature and a Hall-effect sensor with a wheel-mounted magnet to measure wheel RPM. Using the measured tire diameter of **4.61 inches**, I converted live RPM into real-time vehicle speed, reaching a measured top speed of **8.23 mph** during testing.

The telemetry system now provides live:

- Battery Voltage
- Motor Temperature
- Wheel RPM
- Vehicle Speed

Next, I will integrate the BNO055 IMU for pitch and roll, followed by Bluetooth communication and development of the mobile telemetry application.

